

Multiple Choice (10 marks)

1. A reaction that includes energy as one of its reactants is called a(n)
 - (a) endergonic reaction.
 - (b) catabolic reaction.
 - (c) redox reaction.
 - (d) anabolic reaction.
 - (e) exergonic reaction.
2. Explain why simple diffusion is not sufficient to account for the passage of all of the dissolved minerals into the cells of water plants.
 - (a) Simple diffusion can only transport minerals when they are bound to oxygen molecules
 - (b) Simple diffusion cannot occur against a concentration gradient
 - (c) Simple diffusion requires specific carrier proteins for each mineral
 - (d) Simple diffusion only works for organic compounds, not inorganic minerals
 - (e) Simple diffusion can transport minerals in any concentration
3. Which of the following is a second messenger that stimulates release of calcium ions into the cytoplasm?
 - (a) Adenosine triphosphate
 - (b) Nitric oxide
 - (c) Guanosine triphosphate
 - (d) Cyclic GMP
 - (e) Prostaglandins
4. All of the following statements are true EXCEPT
 - (a) insulin decreases storage of glycogen
 - (b) growth hormone stimulates cell division and protein synthesis
 - (c) epinephrine increases blood sugar levels and heart rate
 - (d) thyroxine increases the rate of metabolism
 - (e) vasopressin stimulates water reabsorption in the kidney
5. One group of compounds that may become quite important in birth control is the prostaglandins. What are the advantages of prostaglandins in birth control?

- (a) Prostaglandins are used to enhance mood and have no effect on birth control.
- (b) Prostaglandins increase fertility by promoting ovulation.
- (c) Prostaglandins are able to prevent pregnancy by altering the DNA of the sperm and egg.
- (d) Prostaglandins are known to mediate hormonal action
- (e) Prostaglandins are non-hormonal, reducing the risk of hormonal side effects.

True / False (10 marks)

Answer True or False

- 6. True or False: Crossing-over during meiosis increases the genetic variation in zygotes, which can help estimate mutation rates in populations.
- 7. True or False: All protozoans exhibit a uniform reproductive method, making them easy to categorize within a single phylum.
- 8. True or False: Measuring the physical characteristics of a population can reliably indicate whether it is in genetic equilibrium.
- 9. True or False: Cyanobacteria are prokaryotic organisms that lack nuclei, distinguishing them from green unicellular algae.
- 10. True or False: The animal pole of telolecithal eggs undergoes cleavage more readily than the vegetal pole.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the various methods used to form recombinant DNA, highlighting which processes are essential and which are not, such as the role of translation in this context.
- 12. Discuss the factors that cause electrons to transition between orbitals in an atom, and explain the implications of these changes on their energy and position.

End of Paper